CCC-C-439b
June 22, 1962
SUPERSEDING
Fed. Spec. CCC-C-439a
JULY 27, 1959

FEDERAL SPECIFICATION

CLOTH, COTTON, BUNTING, MERCERIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton bunting primarily intended for use in the manufacture of signal, semaphore, ensign (including U.S. flag), and other type flags.

1.2 Classification.

1.2.1 *Type.* Bunting covered by this specificatian shall be of one type only.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 *Specifications and standards.* The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

P–S–617—Soap, Toilet, Hard, Soft or Sea Water.
CCC–T–191—Textile Test Methods.
PPP–P–51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cotton, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes culmulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Color Association of the United States, Inc.
Standard Color Card of America.

(Copies of this color card may be obtained from the Textile Color Card Association of the United States, Inc., 200 Madison Avenue, New York 16, New York.)

Federal Trade Commission Publication:
Rules and Regulations Under the Textile

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

Fiber Products Identification Act, Effective March 3, 1960.

(Copies may be obtained without charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced. (See 6.3.)

3.2 Material.

3.2.1 *Yarn.* The yarn shall be two-ply in both warp and filling and shall be made from cotton which has been cleaned, carded, and drawn.

3.3 Color. The color shall be as specified. (see 6.1) and shall be produced by such colorants and methods as to meet the fastness requirements of 3.3.2. When the basic colors of the United States Flag are required, they shall match cable numbers 70180 (red), 70-001 (white), and 70075 (blue) of the Standard Color Card of America, as applicable (see 2.2). The red (70180) may be dyed with naphthol dyes and the blue (70075) shall be dyed with vat dyes.

3.3.1 *Matching.* The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7,500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* The finished cloth shall show fastness to crocking, light, water, and salt water and soap equal to or better than the standard sample when tested as specified in 4.4. When a standard sample is not available or is not designated as applicable to colorfastness, the finished cloth shall show fair fastness to crocking and good fastness to light, water, and salt water and soap when tested as specified in 4.4.

3.4 Weave. The weave shall be plain.

3.5 Width. The minimum width, exclusive of selvages, shall be 35-3/4 inches.

3.6 Physical requirements. The finished cloth shall conform to the physical requirements shown in table I when tested as specified in 4.4.

3.7 Finish. The cloth shall be fully mercerized. The pH shall be not less than 6.0 nor more than 9.0 when tested as specified in 4.4.

3.8 Nonfibrous materials. The residual size, finishing and nonfibrous materials shall not exceed 2.0 percent when tested as specified in 4.4.

3.9 Length. Unless otherwise specified, the cloth shall be in one continuous length of not less than 35 yards.

3.10 Fiber identification. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act, Effective March 3, 1960 (see 2.2).

3.11 Workmanship. The bunting shall conform to the quality and grade of product established by this specification. The occurance of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Except when otherwise specified by civil agencies, inspection shall be in accordance with MIL–STD–105, except where otherwise indicated. Defects found during this examination shall be classified in accordance with 4.2.1, 4.2.2, and 4.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on one side and visual defects classified as listed below. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case, only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The inspection level shall be level II of MIL–STD–105 and the acceptable quality levels shall be 1.5 major defects and 6.5 total defects (major and minor combined) per 100 units (yards). The lot size shall be expressed in units of 1 yard. The number of sample pieces over which the sample yardage shall be distributed shall be in accordance with table II. An approximate equal number of yards shall be examined in each piece. No more than one piece shall be selected from any one container.

Yard-by-yard examination

Glossary No. of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
90, 91, 92	Hole, cut or tear - Any (including selvage)	X	
11, 33	Broken or missing yarns:		
	Two or more contiguous, 1/2 inch or more missing	X	
	Single end missing, 2 inches or more		X
	Single pick missing more than 9 inches		X
	Note: A single broken pick regardless of length will not be tallied if it does not result in a clearly visible variation in filling texture.		
78	Smash - Any	X	
12	Crack or open place*	X	
4, 10, 13	Mispick - Resulting in double pick with more than two yarns in shed.	X	
79, 89	Abrasion mark - Resulting in a thin or weak place	X	
34	Float:		
	Multiple more than 1/2 inch in combined length and width dimension.	X	
	Multiple, 1/2 inch or less in combined dimension		X
8, 9	Fine or light filling bar*	X	
2	Coarse or heavy filling bar*		X
38	Slack end - extending more than 9 inches and forming clearly noticeable loops on surface of cloth.		X
5, 27, 77	Slub, slug, gout or kink - More than 1/8 inch in thickness and over 1 inch in length.		X
	Smaller than described above but more than 5 in any 1/4 linear yard.		X
3, 5	Jerked-in filling or slough-off:		
	Three or more additional yarns in shed	X	
	Two additional yarns in shed		X
14	Shade bar or mixed filling bar*	X	
40, 41	Reed mark or wrong draw - More than 18 inches in length*.		X
44, 69, 70, 74, 75, 76 [1]	Spot, stain or streak*		X

* Clearly visible at 6 feet.

[1] Except defects of spot, stain, or streak shall be coun ted regardless of cause, i.e., not restricted to rust, oil, or grease.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

TABLE I. *Physical requirements*

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)	
	Warp	Filling	Warp	Filling
Ounces			*Pounds*	*Pounds*
4.5	31	30	50	50

4.2.2 *Overall examination.* Each defect shall be counted no more than once in each piece. The sample unit for this examination shall be one piece. The sample size (number of pieces) for this examination and the maximum number of defects acceptable shall be in accordance with table II.

Defects

Width not as specified.
Baggy, ridgy or wavy cloth, clearly noticeable.
Selvage curled, doubled, folded, rolled, slack, tight, intermittently or continuously over more than ½ length of piece.
Off shade—not within the established tolerance.
Not uniform in shade—shaded end-to-end, side-to-side, side-to-center, mottled, or poor dye penetration, two-sideness.
Overall uncleanliness.
Objectionable odor.
Neppiness, clearly noticeable throughout.
Not uniformly woven throughout.

4.2.3 *Examination for length.*

4.2.3.1 *Individual pieces.* Each piece in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum, or any gross length found to be more than 2 yards below the gross length marked on the piece ticket shall be considered a defect with respect to length. The sample unit for this examination shall be one piece. The number of pieces selected as a sample (sample size) and the acceptance number shall be in accordance with table II.

4.2.3.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on the piece tickets.

TABLE II. *Sampling for inspection*

Lot size in yards	Sample size in pieces	Acceptance number (defects)
Up to 1,300 [1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 and over	25	0

[1] If the lot contains less than 3 pieces, each piece in the lot shall be examined.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed on the following page shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight	Gross or net weight exceeds requirements.

4.4 Tests. The methods of testing specified in CCC–T–191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 of this specification apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for test purposes shall be 1-1/2 continuous yards full width of finished cloth. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	*Sample size*
800 or less	2
801 to 22,000	3
22,001 and over	5

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements).

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.1 and 6.2).

5.1.1 *Levels A, B, and C.* The cloth put up in flat folded pieces or on rolls shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP–P–51.

TABLE III. *Test methods*

Characteristic	Requirement reference	Method number
Nonfibrous material	3.8	2611
Weight, ounces per square yard	Table I	5041
Yarns per inch	Table I	5050
Breaking strength	Table I	5100
Ply	3.2.1	Visual [1]
Weave	3.4	Visual [1]
pH	3.7	2810 or 2811
Mercerization	3.7	Microscopic examination for swelling of fibers [1]
Colorfastness to:		
Crocking	3.3.2	5651
Light	3.3.2	5660 [2]
Water	3.3.2	5630
Salt water and soap	3.3.2	5632 [3]
Vat dye	3.3	[4]

[1] One determination per sample unit, and results reported as "pass or fail".

[2] In the event of dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulation, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[3] Method 5632, except that the soap shall meet the salt-water soap requirements of P–S–617.

[4] This requirement may be accepted on the basis of contractors certificate of compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1 and 6.2).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP–P–51.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*a*) Color (see 3.3).
(*c*) Minimum length, if other than specified (see 3.9).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 In the preparation of contracts or orders, it should be noted that various options, choices, and alternatives, as indicated in PPP–P–51, may be exercised in the preparation for delivery of this commodity.

6.3 Standard sample. For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids.

6.4 Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Cotton piece goods finished, not otherwise indexed by name:
Carload minimum weight—any quantity.
Motor:
Cotton piece goods, not otherwise indexed:
Motor volume minimum weight—any quantity.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q
Navy—S MC
Air Force—ASD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:30Z
Check the source to verify that this is the current version before use.